A APPENDIX

In this appendix, we provide additional details to the main text, including:

- A.1: Detailed experimental setups, including the prompt template.
- A.2: Full results on GSM8K, GSM-Symbolic, and their variants.
- A.3: Additional results for the distributional performance of several models, similar to the results from Sec. 4.1 in the main text.
- A.4: Additional results for Sec. 4.3, where we studied the impact of question difficulty. We show that fine-tuning on easier tasks does not necessarily improve performance on more difficult tasks.
- A.5: A more comprehensive discussion and analysis of performance for OpenAI o1-mini and o1-preview models.

A.1 DETAILED EXPERIMENTAL SETUP

In this work, all reported evaluations results use 8-shots with chain-of-thought prompting. We use the following prompt format:

Evaluation Prompt Format

```
// preamble or system instruction
As an expert problem solver, solve step by step the following mathematical questions.

// shot-1
Q: {{question}}
A: Let's think step by step.  {{solution}}.  The final answer is {{final answer}}.
.
.
.
// shot 8
Q: {{question}}
A: Let's think step by step.  {{solution}}.  The final answer is {{final answer}}.

// target question
Q: {{question}}
A: Let's think step by step.
```

Figure 9: The prompt format used for evaluations.

Except for the last experiment in Sec. 4.4, we use the original 8 shots from GSM8K. In addition, we allow the models to generate until either their context size limit is reached, they generate one of the end-of-response tokens such as '</s>' or '<|endoftext|>', or they finish answering the current question and move on to generating the next question, indicated by another 'Q:' generation.

Finally, we note that in all experiments we use greedy decoding to generate responses from models, with one exception: currently, the available APIs for "o1-mini" and "o1-preview" models do not allow controlling the decoding strategy, and it seems that at the time of writing, these models do not perform greedy decoding, as responses to the same prompt change.

A.2 FULL RESULTS

In Tab. 1, we present the comprehensive performance results of various models, including Gemma (Mesnard et al., 2024), Gemma2 (Rivière et al., 2024), Phi (Abdin et al., 2024), Mistral (Jiang et al., 2023), Llama3 (Dubey et al., 2024), GPT-4o (OpenAI, 2023), and the o1 (OpenAI, 2024) series, on GSM8K and its different variants, GSM-Symbolic.

We report two sets of results for GSM8K: the first column indicates the accuracy on the *full* test set of GSM8K (comprising 1,319 examples), while the second column shows the accuracy on a subset of 100 questions from the GSM8K test set, which we randomly selected to generate GSM-Symbolic

templates. It is noteworthy that the performance levels across both sets are very similar, with no significant differences observed.

Table 1: Full 8-shot results of all models on GSM8Kand different variants of GSM−Symbolic.

Model	GSM8K (Full)	GSM8K (100)	Symbolic-M1	Symbolic	Symbolic-P1	Symbolic-P2	Symbolic-NoOp
Gemma2b	12.1	11.0	24.5 (± 3.85)	8.2 (± 2.21)	3.6 (± 2.13)	1.5 (± 1.63)	4.7 (± 1.99)
Gemma2b-it	12.1	11.0	16.2 (± 3.28)	8.2 (± 2.21)	1.5 (± 1.49)	1.5 (± 1.63)	4.1 (± 2.48)
Gemma-7b	53.8	50.0	34.1 (± 4.41)	25.6 (± 3.25)	26.0 (± 5.30)	3.1 (± 1.92)	8.7 (± 2.71)
Gemma-7b-it	29.3	33.0	34.1 (± 4.41)	25.6 (± 3.25)	6.0 (± 3.38)	3.1 (± 1.92)	8.7 (± 2.71)
Gemma2-2b	47.5	46.0	57.2 (± 3.40)	40.1 (± 3.04)	19.5 (± 3.89)	1.3 (± 1.37)	8.8 (± 4.12)
Gemma2-2b-it	47.5	46.0	57.2 (± 3.40)	40.1 (± 3.04)	19.5 (± 3.89)	4.5 (± 1.94)	15.7 (± 3.97)
Gemma2-9b	85.3	87.0	71.2 (± 2.81)	79.1 (± 2.99)	44.0 (± 5.69)	41.8 (± 6.00)	22.3 (± 5.11)
Gemma2-9b-it	85.3	87.0	84.4 (± 2.36)	79.1 (± 2.99)	68.1 (± 4.77)	41.8 (± 6.00)	22.3 (± 5.11)
Gemma2-27b-it	89.7	92.0	90.2 (± 1.86)	88.3 (± 2.56)	80.7 (± 4.07)	63.4 (± 4.14)	30.0 (± 3.39)
Phi-2	56.0	53.0	53.0 (± 3.10)	41.4 (± 3.56)	23.3 (± 4.07)	8.9 (± 3.33)	11.2 (± 3.51)
Phi-3-mini-128k-instruct	83.7	85.0	85.9 (± 2.44)	80.7 (± 2.94)	63.4 (± 5.63)	37.5 (± 5.76)	18.0 (± 3.83)
Phi-3-small-128k-instruct	88.5	89.0	86.4 (± 1.95)	83.7 (± 2.65)	72.0 (± 3.65)	50.7 (± 4.99)	24.5 (± 4.81)
Phi-3-medium-128k-instruct	87.3	89.0	89.6 (± 1.65)	82.5 (± 2.86)	75.8 (± 3.89)	53.1 (± 4.80)	29.4 (± 4.18)
Phi-3.5-mini-instruct	84.9	88.0	87.6 (± 1.98)	82.1 (± 3.38)	64.8 (± 5.43)	44.8 (± 6.32)	22.4 (± 4.03)
Mistral-7b-v0.1	44.5	48.0	55.4 (± 3.18)	41.1 (± 3.36)	17.4 (± 4.82)	5.5 (± 2.55)	16.2 (± 4.43)
Mistral-7b-instruct-v0.1	39.7	42.0	44.9 (± 4.29)	30.5 (± 3.47)	13.1 (± 3.51)	4.0 (± 2.24)	10.1 (± 3.42)
Mistral-7b-v0.3	40.6	44.0	54.0 (± 2.95)	40.0 (± 4.43)	15.6 (± 4.02)	3.9 (± 2.31)	16.7 (± 4.26)
Mistral-7b-instruct-v0.3	56.2	56.0	62.3 (± 2.68)	50.0 (± 3.49)	24.5 (± 4.34)	10.8 (± 3.60)	15.9 (± 4.44)
Mathstral-7b-v0.1	80.1	80.0	82.9 (± 2.87)	74.0 (± 3.49)	57.4 (± 5.20)	35.5 (± 5.07)	20.4 (± 3.58)
Llama3-8b	55.8	61.0	79.5 (± 3.62)	74.6 (± 2.94)	53.8 (± 4.54)	12.3 (± 3.43)	18.6 (± 3.86)
Llama3-8b-instruct	76.0	74.0	79.5 (± 3.62)	74.6 (± 2.94)	53.8 (± 4.54)	28.3 (± 4.37)	18.6 (± 3.86)
GPT-4o-mini	94.2	95.0	92.5 (± 1.63)	91.7 (± 2.02)	81.1 (± 3.05)	72.4 (± 4.57)	54.1 (± 3.85)
GPT-4o	95.2	95.0	94.4 (± 1.62)	94.9 (± 1.87)	93.9 (± 2.59)	88.0 (± 3.43)	63.1 (± 4.53)
o1-mini	95.1	93.0	94.9 (± 1.49)	94.5 (± 1.58)	94.3 (± 2.57)	89.1 (± 3.56)	66.0 (± 4.60)
o1-preview	94.9	96.0	93.6 (± 1.68)	92.7 (± 1.82)	95.4 (± 1.72)	94.0 (± 2.38)	77.4 (± 3.84)

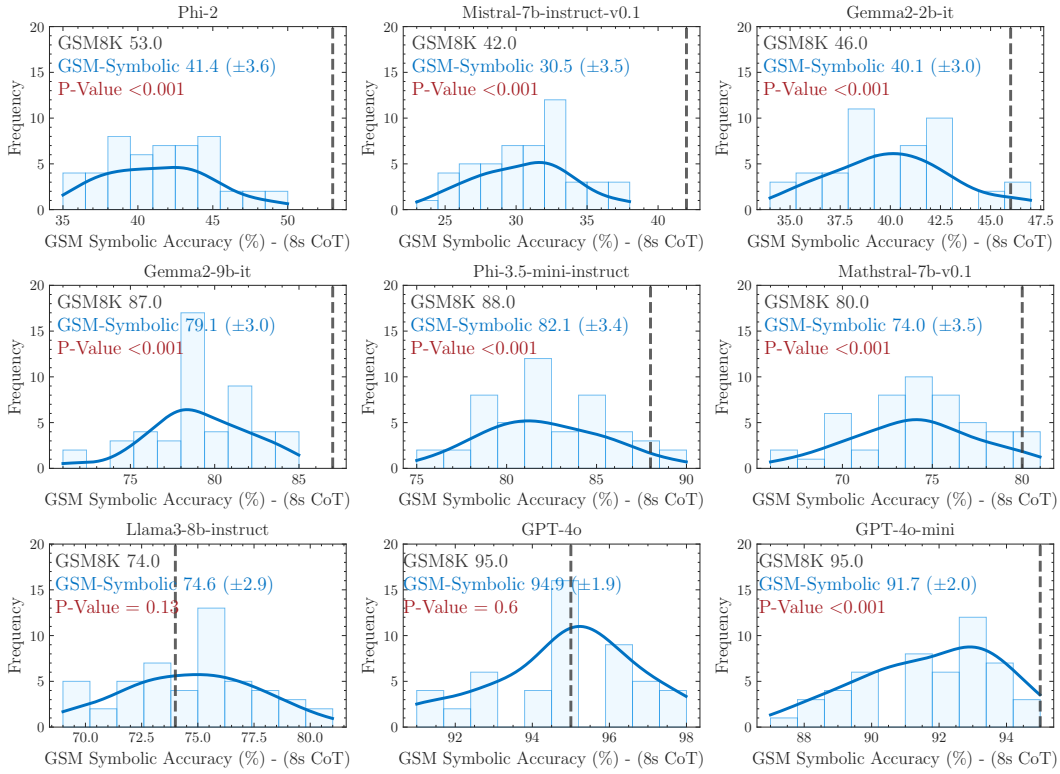

Figure 10: Additional results on performance variation on GSM−Symbolic.

A.3 ADDITIONAL RESULTS ON GSM-SYMBOLIC PERFORMANCE DISTRIBUTIONS

In section 4.1, we have presented results for several models in Fig. 2. Here, we provide additional results showing the performance on GSM-Symbolic for other models also have high variance. Moreover, these models correspond to highest drop.

Another important question regarding our results is measuring the statistical significance[3]. In this work, we calculate statistical significance using the one-sample t-test with two-sided alternative hypothesis to determine whether the 50 different performance results on GSM-Symbolic differ from the original 100 questions from GSM8K (i.e., the null hypothesis)[4]. As we can see in Fig. 10, for an overwhelming majority of models (except Llama3-8B and GPT-4o), the results are statistically significant.

Thus far, we have assessed the probability of obtaining the 50 GSM-Symbolic scores under the assumption that the population mean equals the accuracy on the original 100 questions from GSM8K. However, Ivanova et al. (2025) argue that a one-sample t-test is inappropriate because the evaluation involves two datasets, and that a two-sample z-test would therefore be more suitable.

Overall, we do not believe that extensive statistical testing adds significant value the main focus of this paper. We discuss our position further in Sec. B. Nonetheless, in the interest of transparency, we clearly state our assumptions and present alternative viewpoints. Thus, if one disagrees with our assumptions and accepts the assumptions of Ivanova et al. (2025), our results would likewise be statistically non-significant, as those authors have shown.

A.4 ABLATION: DOES FINE-TUNING ON EASIER TASKS HELP WITH MORE DIFFICULT TASKS?

In Sec. 4.3, we observed that the performance on GSM-P2 is significantly lower than the performance on GSM-P1. We also argued that it is unlikely that additional fine-tuning or including shots from GSM-P1 would be beneficial. Here, in Fig. 11a, we show that including shots from GSM-P1 does not improve performance compared to the results where shots come solely from GSM8K.

Moreover, in Fig. 11b, we demonstrate that fine-tuning Phi-3.5 on GSM-P1 slightly improves performance on GSM-P1 while decreasing performance on GSM-P2. We have used a set of 50 templates from GSM-P1, separate from the test templates, and generated 10000 examples for finetuning training set.

Overall, while this direction warrants further research, current results suggest that scaling training data will not be helpful in improving the reasoning capabilities of language models.

A.5 RESULTS ON O1-PREVIEW AND O1-MINI

The recently released o1-preview and o1-mini models (OpenAI, 2024) have demonstrated strong performance on various reasoning and knowledge-based benchmarks. As observed in Tab. 1, the mean of their performance distribution is significantly higher than that of other open models.

In Fig. 12 (top), we illustrate that both models exhibit non-negligible performance variation. When the difficulty level is altered, o1-mini follows a similar pattern to other open models: as the difficulty increases, performance decreases and variance increases.

The o1-preview model demonstrates robust performance across all levels of difficulty, as indicated by the closeness of all distributions. However, it is important to note that both o1-preview and o1-mini experience a significant performance drop on GSM-NoOp . In Fig. 13, we illustrate that

[3]This is a very complicated topic, as the notion of statistical significance depends on many assumptions about the properties of data. For instance, one could view the overall performance of models as a Bernoulli trial, with model accuracy representing the probability of success. However, this requires an i.i.d. assumption on the questions and the model accuracy for each question, which may not necessarily hold and needs further investigation.

[4]In general, the t-test assumes normality, which may not necessarily hold for evaluation results. However, using a standard normality test based on the skewness and kurtosis of samples, we have verified that all distributions in Fig. 10 pass the normality test (p-value of the normality test = 0.1).